

Azure Virtual Machine Extension Standardisation

Problem Statement:-

Azure Virtual Machines may have different configurations when their extensions are installed manually or at different times. Some VMs may not have the required monitoring or security agents, while others may have failed or conflicting extensions. These problems can create security gaps, reduce monitoring coverage, and consume unnecessary CPU and memory resources.

The problem is to standardise Azure VM extensions by checking whether the required monitoring and security agents are installed and functioning correctly. Missing or failed extensions should be detected and remediated, while unnecessary or conflicting agents should also be identified.

Abstract:-

Azure Virtual Machine Extension Standardisation is a process used to ensure that all Azure Virtual Machines have the required monitoring and security extensions installed and properly configured. In large cloud environments, manually managing VM extensions can lead to missing extensions, failed installations, and inconsistent configurations. The proposed approach checks each VM for the required extensions, identifies missing or failed extensions, and performs remediation when necessary. It also helps detect multiple or conflicting agents that may consume excessive system resources. This improves VM security, monitoring, consistency, and overall cloud management.

Use Cases:-

Use Case 1: Ensure Required Extensions

Check every Azure Virtual Machine and verify that the required monitoring and security agents are installed.

Use Case 2: Detect Missing Extensions

Identify VMs where one or more required extensions are missing.

Use Case 3: Remediate Missing Extensions

Install or repair the required extension when it is missing or has failed.

Use Case 4: Detect Extension Conflicts

Identify duplicate or conflicting monitoring and security agents that may consume unnecessary CPU, memory, or other resources.

Use Case 5: Maintain Standard Configuration

Ensure that all Azure VMs follow the same approved extension configuration.

Step-by-step procedure:-

Step 1: Start the process.

Step 2: Collect the list of Azure Virtual Machines.

Step 3: Check the extensions installed on each VM.

Step 4: Compare the installed extensions with the required standard extensions.

Step 5: If all required extensions are present, check whether they are running correctly.

Step 6: If an extension is missing, identify the missing extension.

Step 7: Install or remediate the missing/failed extension.

Step 8: Check whether multiple monitoring or security agents are conflicting.

Step 9: If conflicts exist, take corrective action such as removing unnecessary agents.

Step 10: Mark the VM as Compliant when the required extensions are installed and working correctly.

Step 11: Generate the final compliance report.

Step 12: End the process.